The Bombay Salesian Society's

DON BOSCO INSTITUTE OF TECHNOLOGY

Premier Automobiles Road, Kurla West, Mumbai - 400070

IEEE - DBIT

Report on – COLOHACKS (HARDWARE)

Title: Colohacks (Hackathon - Hardware)
Date: 24th & 25th (overnight 24hrs)
Time: 10 a.m. – 4 p.m.
Venue: B – WING 1ST FLOOR EXTC DEPARTMENT

Target Audience: Open For All

No. of Participants Present: 92

No. of Girl Participants Present: 21

No. of Boy Participants Present: 71

Faculty Coordinator: Ms Freda Carvalho

Objectives:

- Encourage innovation and practical problem-solving through hardware-based projects
- Provide a platform for students from multiple colleges to collaborate and compete
- Promote hands-on learning and technical skill development in real-time
- Foster teamwork, time management, and endurance in a high-pressure environment
- Enable participants to transform ideas into working prototypes
- Facilitate continuous mentorship and guidance to improve project quality
- Conduct a structured evaluation process through multiple judging rounds

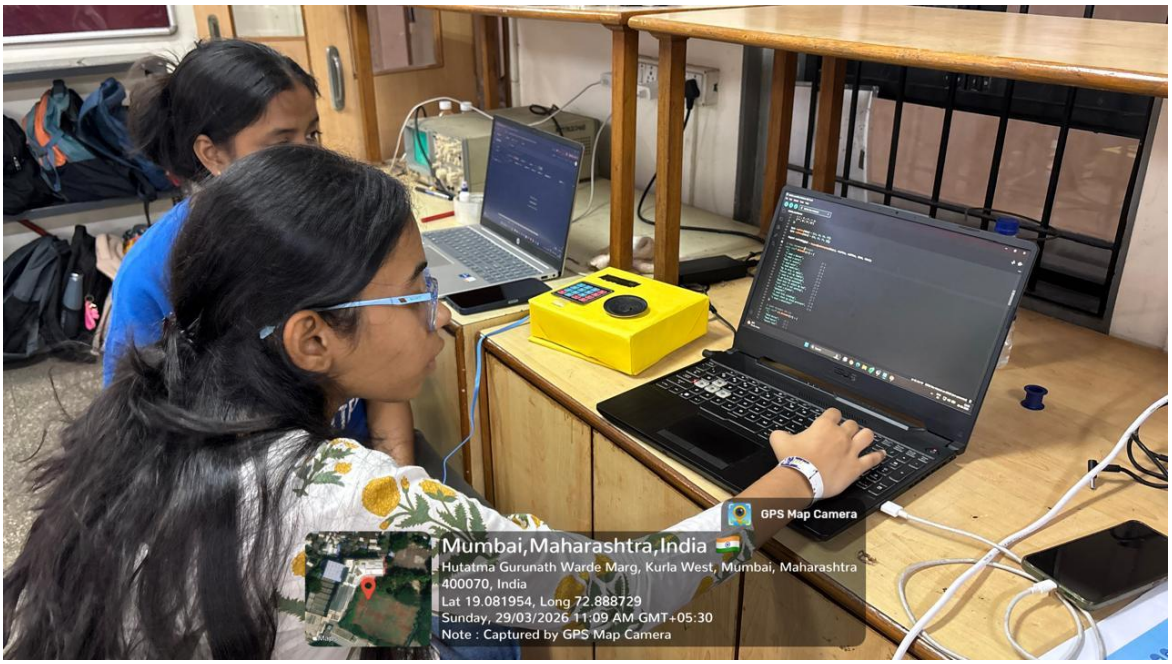
Outcomes:

- Participants developed functional hardware prototypes within a limited time frame
- Improved technical, design, and implementation skills among students
- Enhanced ability to work efficiently under pressure and tight deadlines
- Strengthened team coordination and task distribution strategies
- Exposure to real-world problem-solving and innovation challenges
- Identification of top 10 high-performing teams through rigorous evaluation
- Boosted confidence, creativity, and competitive spirit among participants

Detailed Report:

The 24-hour Hardware Hackathon organised by IEEE DBIT under Colosseum 2026 was a highly intensive and innovation-driven event conducted from 10:00 AM on 24th March to 4:00 PM on 25th March, bringing together participants from multiple colleges to design and develop real-time hardware solutions addressing practical, real-world problems. The event began with a rigorous selection process conducted through Unstop, where teams were required to submit a detailed presentation in the form of a PPT along with a video explanation of their proposed ideas, out of which only 23 teams were shortlisted for the final offline hackathon. The hackathon officially commenced at 10:00 AM, with all selected teams beginning work simultaneously on their respective projects, showcasing immense dedication, creativity, and technical expertise. Participants worked continuously throughout the 24-hour duration, effectively managing time and energy by taking short breaks in shifts while ensuring consistent progress in development and implementation. Continuous mentorship was provided throughout the event, where experienced mentors guided teams in refining their concepts, troubleshooting technical challenges, and improving the feasibility and scalability of their solutions. The evaluation process was structured into three rounds of judging to ensure a fair and comprehensive assessment, with the first round focusing on idea validation and initial progress, the second round evaluating implementation and working prototypes, and the final round determining the top 10 teams based on innovation, execution, and real-world applicability. The top-performing teams presented highly impactful and diverse solutions across domains such as healthcare, smart cities, environmental sustainability, and AI-driven systems. Several healthcare-focused innovations stood out, including a portable IoT-enabled life-saving kit designed for emergency and rural healthcare scenarios, a distributed mini ICU system aimed at providing affordable and continuous patient monitoring beyond hospitals, and a smart rehabilitation glove using AI and servo-assisted mechanisms to support physiotherapy and stroke recovery. Additionally, an IoT-based smart inhaler was developed to monitor usage patterns, prevent overuse, and provide real-time tracking through cloud integration. In the domain of smart cities and environmental monitoring, teams proposed innovative solutions such as an environmental monitoring station capable of tracking multiple air quality parameters using IoT and long-range communication technologies, and a water quality monitoring system designed to provide real-time analysis of parameters like turbidity, pH, and TDS for safe drinking water. AI-based systems also played a significant role, with one team developing an adaptive traffic management system using computer vision and real-time analytics to optimise traffic flow and reduce congestion and accidents. Another notable innovation was a multimodal neuro-health monitoring system that integrates brain signals, heart activity, and environmental data to detect early neurological imbalances and provide deeper health insights. Overall, the hackathon successfully created a highly competitive and collaborative environment that encouraged innovation, critical thinking, and teamwork under pressure, while providing participants with valuable hands-on experience and exposure to real-world engineering challenges. The event concluded with the selection of the top 10 teams, marking it as a successful and impactful initiative that highlighted the potential of student-driven innovation in solving pressing societal issues.

Snapshot of the Event:



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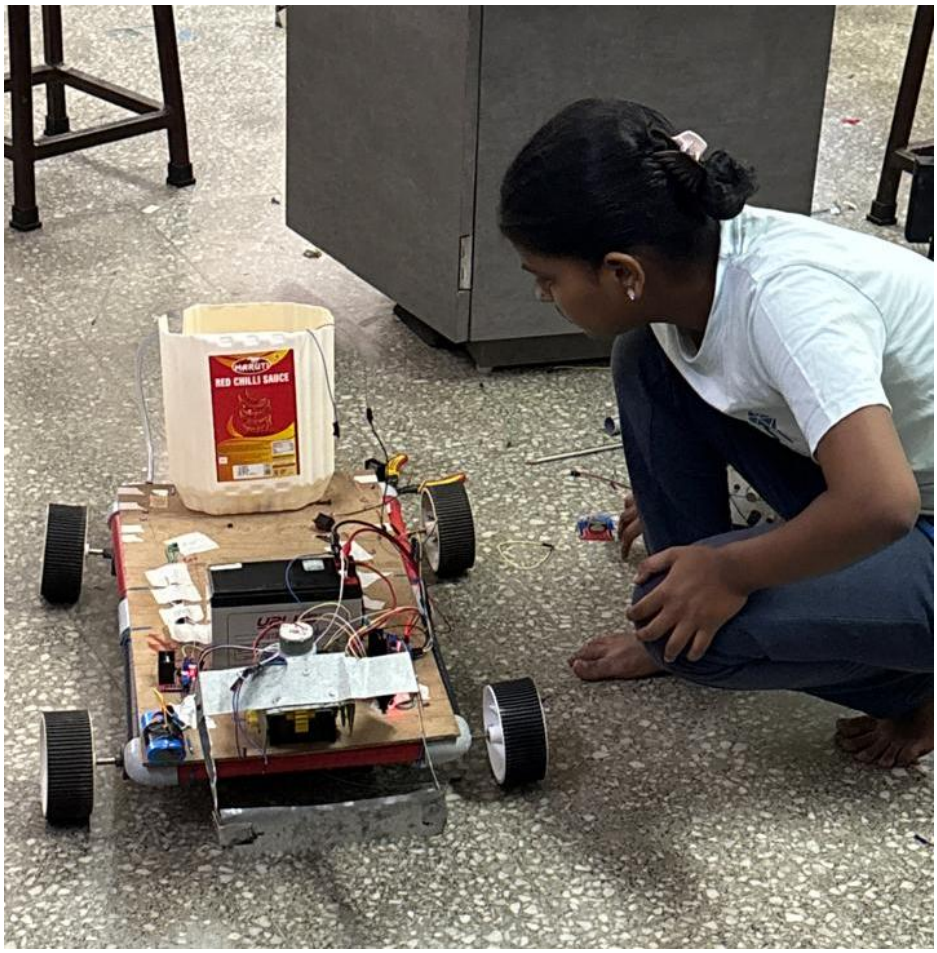

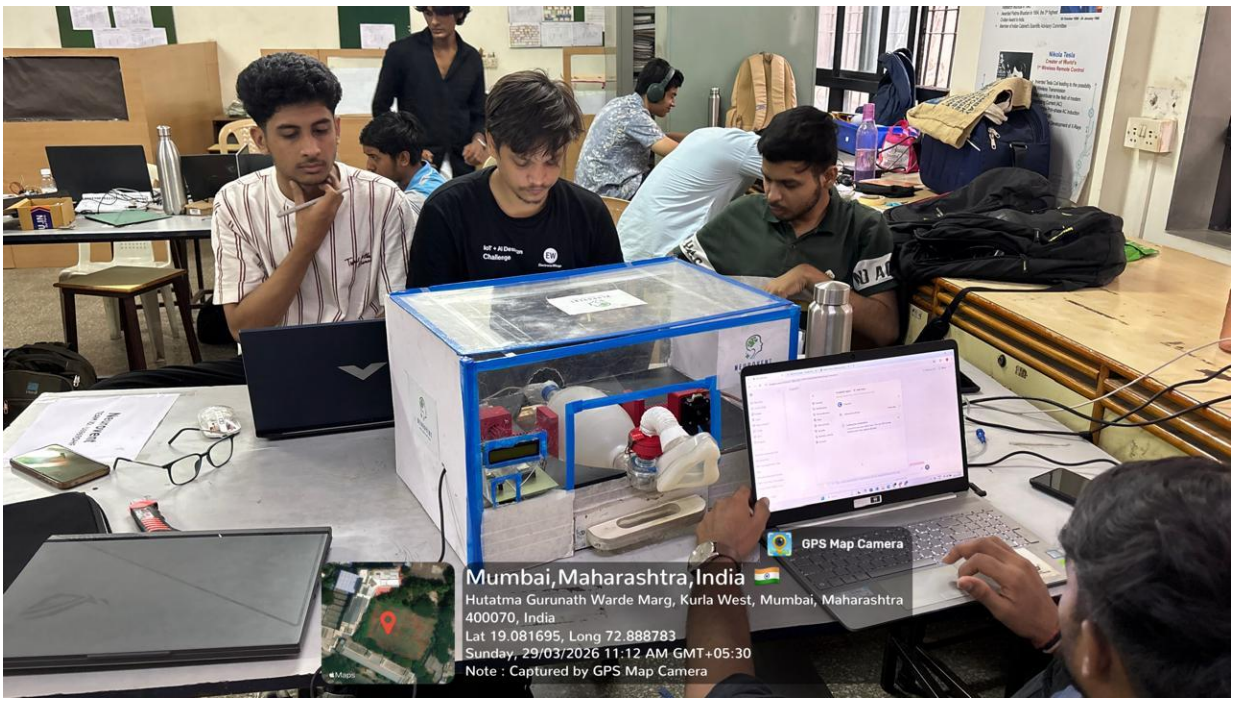

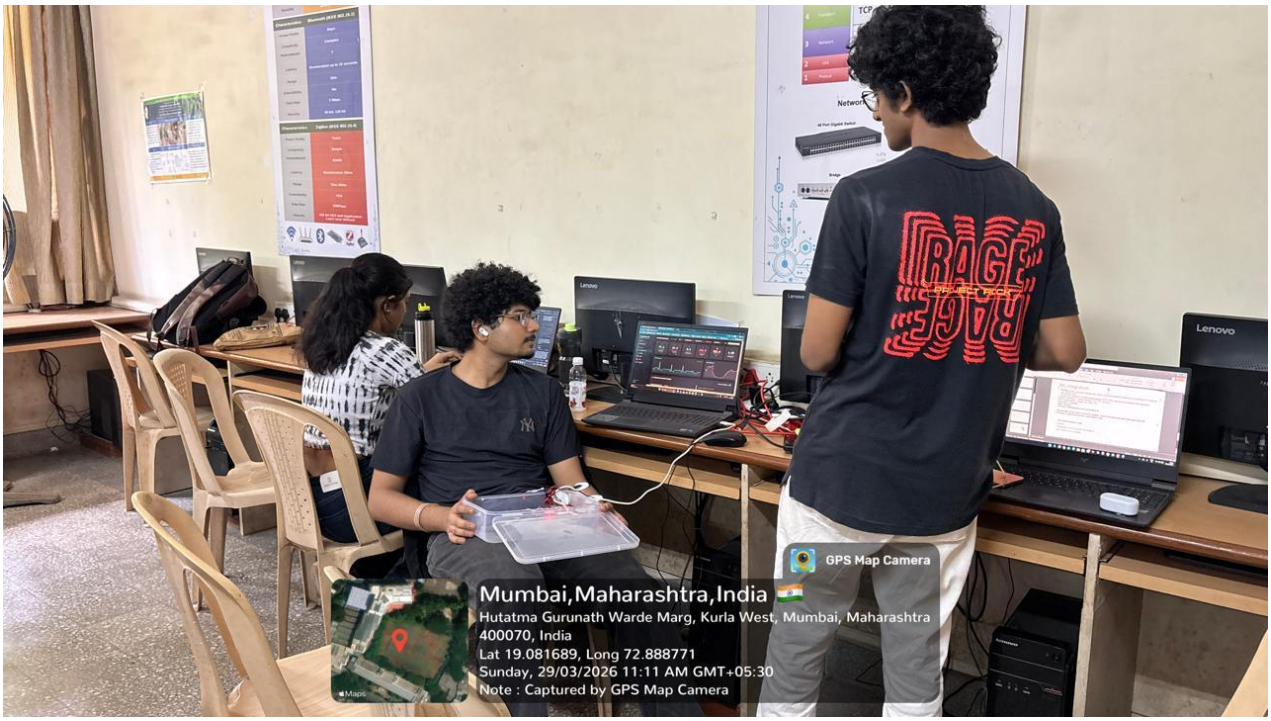

Feedback Analysis: ***(Justified, Text in 12 point font size, Times Roman New, 1.15 line spacing)***

Feedback should include Google Form responses in terms of bar graphs or pie charts.

Event Poster:

Social Media Links:

https://www.instagram.com/ieee_dbit/

https://www.instagram.com/colosseumdbit?igsh=NDN4bWJ6M2c3Zmtn&utm_source=qr

Registration Details:

In case the event is open for non-DBIT students:

No. of DBIT Students: 36
No. of non-DBIT students: 56

Sr. no.	Team Name	Candidate's Name	Specialization
1.	Susx	Jagdish Dilip Wagh	Artificial Intelligence And Data Science
2.	Susx	Rohit Krishna Chavan	Computer Science And Engineering
3.	Susx	Ibrahim Abdul Razzak Shaikh	Computer Science And Engineering
4.	Susx	Ayush Sharban Thakur	Artificial Intelligence And Machine Learning Engineering
5.	Bolts & Bots	Hrishikesh Nilesh Dhume	Electronics And Communication
6.	Bolts & Bots	Anmol Katiyar	Electronics And Communication
7.	Bolts & Bots	Soumil Chawan	Electronics And Communication
8.	Bolts & Bots	Archit Langde	Electronics And Communication
9.	Theinnovationarc	Sanjana Jagtap	Computer Science And Engineering
10.	Theinnovationarc	Sejal Gupta	Cyber Security
11.	Theinnovationarc	Prince Gupta	Artificial Intelligence And Machine Learning Engineering
12.	Theinnovationarc	Divyansh Pandey	Electrical Engineering
13.	Code Red	Aayush Manoj Pataskar	Electronics And Communication
14.	Code Red	Anvit Mulik	Electronics And Communication
15.	Code Red	Trushna Vilas Hadawale	Electronics And Communication
16.	Code Red	Disha Vora	Electronics And Communication
17.	Ohm My God	Harsh Kumar	Electronics And Communication
18.	Ohm My God	Shravan Tupe	Electronics And Communication
19.	Ohm My God	Vedant Irawati	Electronics And Communication
20.	Ohm My God	Tanisha Kothari	Electronics And Communication
21.	Wavex	Harsh Dhamane	Electronics And Communication
22.	Wavex	Hemanshu Bharambe	Electronics And Communication
23.	Wavex	Anushka Daple	Electronics And Communication
24.	Wavex	Ananya Pawar	Electronics And Communication

25.	Iotians	Om Gupta	Computer Science And Engineering
26.	Iotians	Arushi Shukla	Computer Science And Engineering
27.	Iotians	Rudra Raorane	Computer Science And Engineering
28.	Iotians	Nivedhithaa Selvakumar	Computer Science And Engineering
29.	Technojunkies	Shreya Manoj	Electronics And Communication
30.	Technojunkies	Kshyanika Behera	Electronics And Communication
31.	Technojunkies	Harini Mekala	Electronics And Communication
32.	Technojunkies	Pallavi Gali	Electronics And Communication
33.	Team Trishul	Fanindra Vaishnav	Computer Science And Engineering
34.	Team Trishul	Sparsh Singh	Electronics And Communication
35.	Team Trishul	Shubhank Kela	Electronics And Communication
36.	Hivemind	Sanmeet Dey	Robotics And Automation Engineering
37.	Hivemind	Anurag Rai	Computer Science And Engineering
38.	Hivemind	Mohammad Haris Khan	Robotics And Automation Engineering
39.	Techtribe	Harsh Rajesh Sankhe	Computer Science And Engineering
40.	Techtribe	Sneha Kaidan	Mechanical Engineering
41.	Neurovent	Ayush Brikeshkumar Pandey	Electronics And Communication
42.	Neurovent	Shivam Singh	Electronics And Communication
43.	Neurovent	Pradip Kumar	Electronics And Communication
44.	Neurovent	Taha Sayyad	Electronics And Communication
45.	Krishi Mitra	Satyam Chhatthu Mali	Electrical Engineering
46.	Krishi Mitra	Mohammad Ayaz Khan	Electrical Engineering
47.	Krishi Mitra	Siddhant Sachin Kadam	Electronics And Communication
48.	Krishi Mitra	Mandar Khadsan	Electrical Engineering
49.	Hackronauts	Abdul Qadeer Mujahid	Information Technology
50.	Hackronauts	Areeb Patel	Computer Science And Engineering
51.	Hackronauts	Hassaan Shaikh	Computer Science And Engineering
52.	Hackronauts	Haris Dhukka	Computer Science And Engineering
53.	Mistreated Devs	Karanjawane Parth	Electronics And Communication
54.	Mistreated Devs	Manav Kothari	Electronics And Communication
55.	Mistreated Devs	Tanish Nagarkar	Electronics And Communication
56.	Mistreated Devs	Yaswanth	Electronics And Communication
57.	Piston Power	Om Sanap	Mechanical Engineering
58.	Piston Power	Tanish Kunder	Mechanical Engineering
59.	Piston Power	Mustafa Pathan	Mechanical Engineering

60.	Piston Power	Aditya Parab	Mechanical Engineering
61.	Roboshake	Jaydatt Madan Sawant	Mechanical Engineering
62.	Roboshake	Ashutosh Mishra	Mechanical Engineering
63.	Roboshake	Kartik Nyalapelli	Electronics And Communication
64.	Voltmesh	Yashvi Jain	Electronics And Communication
65.	Voltmesh	Anu Saha	Electronics And Communication
66.	Voltmesh	Netra Hande	Electronics And Communication
67.	Voltmesh	Vipaak Gaikwad	Electronics And Communication
68.	Rjmask	Mrunmai Paktekar	Electronics And Communication
69.	Rjmask	Blossom Mascarenhas	Electronics And Communication
70.	Rjmask	Raman Sharma	Electronics And Communication
71.	Powerpuff	Neel Pujari	Electronics And Communication
72.	Powerpuff	Singh Priyanshukumar Ramasare	Electronics And Communication
73.	Powerpuff	Nisha Singh	Electronics And Communication
74.	Powerpuff	Onkar Lambade	Electronics And Communication
75.	3rd WHEEL	Jaeden Dsouza	Mechanical Engineering
76.	3rd WHEEL	Sumeet Gupta	Mechanical Engineering
77.	3rd WHEEL	Atharva Chache	Mechanical Engineering
78.	Smartdevs	Niraj Vikas Mahajan	Electronics And Communication
79.	Smartdevs	ADNAN QURESHI	Electronics And Communication
80.	Smartdevs	Aakib Patel	Electronics And Communication
81.	Smartdevs	Fayyaz Shaikh	Electronics And Communication
82.	Team404	Sahil Shingre	Electronics And Communication
83.	Team404	Sumit Desai	Electronics And Communication
84.	Team404	Aditya Thakur	Electronics And Communication
85.	Team404	KRITIKUMAR UMESH PAL	Electronics And Communication
86.	The Pathfinder	Sagnik Ulvekar	Mechanical Engineering
87.	The Pathfinder	Sahil Toskar	Mechanical Engineering
88.	Eterna Hackers	Kapil Labde	Electronics And Communication
89.	Eterna Hackers	Om Shashikant Patil	Computer Science And Engineering
90.	Eterna Hackers	Samiksha Gaonkar	Electronics And Communication
91.	Eterna Hackers	Umair Shaikh	Electronics And Communication

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Report Prepared By: IEEE-DBIT

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Report Approved By: IEEE - DBIT

Name of the Faculty: Ms. Freda Carvalho

Post of the Faculty: IEEE-DBIT Student Branch Counsellor